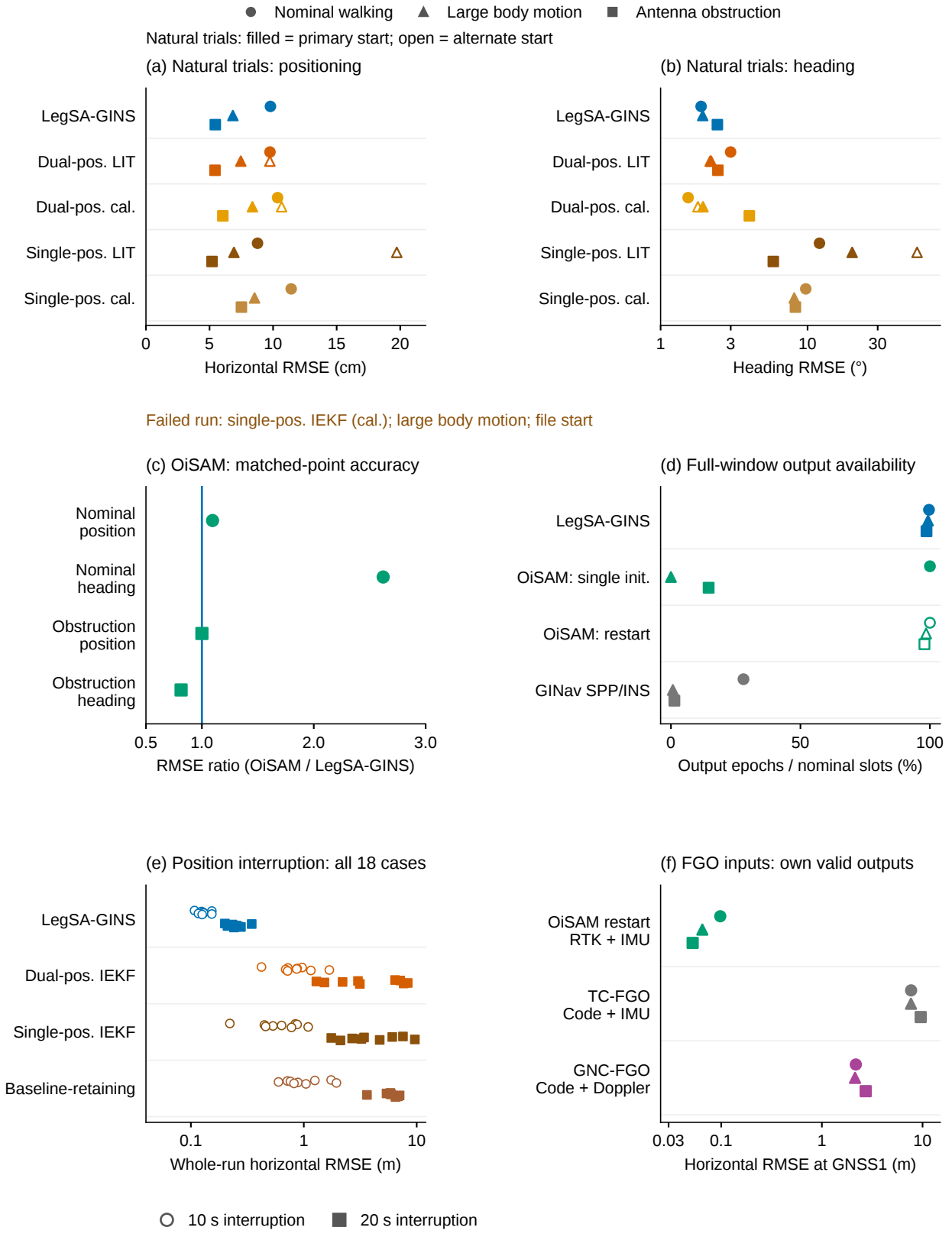
Nominal walking
Large body motion
Antenna obstruction
Natural trials: filled = primary start; open = alternate start
(a) Natural trials: positioning
LegSA-GINS
Dual-pos. LIT
Dual-pos. cal.
Single-pos. LIT
Single-pos. cal.
0
5
10
15
20
Horizontal RMSE (cm)
(b) Natural trials: heading
LegSA-GINS
Dual-pos. LIT
Dual-pos. cal.
Single-pos. LIT
Single-pos. cal.
1
3
10
30
Heading RMSE (°)
Failed run: single-pos. IEKF (cal.); large body motion; file start
(c) OiSAM: matched-point accuracy
Nominal position
Nominal heading
Obstruction position
Obstruction heading
0.5
1.0
2.0
3.0
RMSE ratio (OiSAM / LegSA-GINS)
(d) Full-window output availability
LegSA-GINS
OiSAM: single init.
OiSAM: restart
GINav SPP/INS
0
50
100
Output epochs / nominal slots (%)
(e) Position interruption: all 18 cases
LegSA-GINS
Dual-pos. IEKF
Single-pos. IEKF
Baseline-retaining
0.1
1
10
Whole-run horizontal RMSE (m)
(f) FGO inputs: own valid outputs
OiSAM restart
RTK + IMU
TC-FGO
Code + IMU
GNC-FGO
Code + Doppler
0.03
0.1
1
10
Horizontal RMSE at GNSS1 (m)
10 s interruption
20 s interruption